

Sales Performance Analysis

Using Power BI

Project Objective:

The objective of this project is to analyze sales performance across different categories ,sub-categories, cities,and states using Power BI. The dashboard helps stakeholders monitor sales, profit, targets, quantity sold and order distribution to identify business trends and make data-driven decisions.

Dataset Description:

This dataset contains transactional sales data used to evaluate business performance across products, categories, locations and time periods. The data supports analysis of sales trends, target achievement, profitability, customer purchasing behavior and regional performance, multiple related tables were integrated in power BI to create an interactive dashboard using data modeling and DAX calculations.

The dataset enables:

- Monitoring overall Sales, Profit and Order Volume
- Comparing Actual Sales vs Target Sales
- Identifying high-performing categories and sub-categories
- Tracking monthly sales trends
- Analyzing geographical sales distribution
- Measuring profit margins and business performance indicators

Dataset Tables:

1. **List of Orders** - Contains order details and customer location information.
2. **Order Details** - Includes product sales transaction s and quantities.
3. **Sales Target** - Stores target values for performance comparison.
4. **Order Profit** - Contains profit values for profitability analysis.

Key Business Metrics Generated:

- Total Sales
- Total Profit
- Total Order s
- Profit Margin (%)

- Target Achievement (%)
- Monthly Sales Trend
- Category & Sub-category performance
- City/State-wise Sales Distribution

Column Description:

Column Name	Description
Order ID	Unique identifier for each order
Order Date	Date of purchase
Customer Name	Name of the customer
State	State where the order was placed
City	City where the order was placed
Category	Product category (Electronics, Clothing, Furniture)
Sub-Category	Product sub-category
Quantity	Number of products sold
Amount / Sales	Revenue generated from order
Profit	Profit earned from the sale
Profit Margin	Percentage of profit over sales
Target Sales	Expected sales target
Target	Sales target assigned
Month	Sales month for trend analysis

Data cleaning & transformation:

Data Cleaning:

- Imported datasets into Power BI Power Query Editor.
- Checked and removed duplicate records.
- Identified and handled missing/null values.
- Corrected data types for columns (Text, Number, Date, Currency)

- Renamed columns for better readability and consistency.
- Removed unnecessary columns to improve model performance.
- Standardized category and sub-category values to avoid inconsistencies.

Data Transformation:

- Merged related tables using **Order ID** and category fields.
- Created relationships between:
 - **List of Orders** → **Order Details**
 - **Order Details** → **Sales Target**
 - **Order Details** → **Order Profit**
- Created calculated measures using **DAX**, including:
 - **Total Sales** = SUM(Amount)
 - **Total Profit** = SUM(Profit)
 - **Profit Margin** = DIVIDE(Total Profit, Total Sales, 0)
 - **Target Achievement %**
- Generated date-based fields for **monthly sales trend analysis**.
- Created KPIs to compare **Actual Sales vs Target Sales**.
- Applied aggregations and filters for interactive reporting.
- Built hierarchies for **Category** → **Sub-Category** and **State** → **City** analysis.

Visualization and Insights

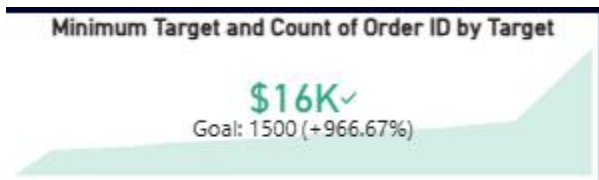
(Attach Screenshot of Visualizations)

1. KPI Cards

1. Monthly Sales and Total Sales by Profit



2. Minimum Target and Count of Order ID by Target



3. Total Target and Total Sales by Sub-Category



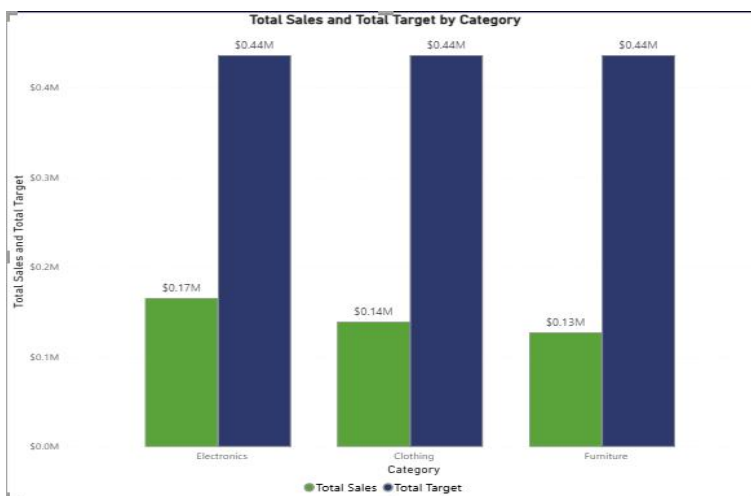
2. Matrix / Table Visual

1. Quantity, Clothing, Electronics, Furniture, Total

Quantity	Clothing	Electronics	Furniture	Total
1	\$174,000	\$129,000	\$132,900	\$435,900
2	\$174,000	\$129,000	\$132,900	\$435,900
3	\$174,000	\$129,000	\$132,900	\$435,900
4	\$174,000	\$129,000	\$132,900	\$435,900
5	\$174,000	\$129,000	\$132,900	\$435,900
6	\$174,000	\$129,000	\$132,900	\$435,900
7	\$174,000	\$129,000	\$132,900	\$435,900
8	\$174,000	\$129,000	\$132,900	\$435,900
Total	\$174,000	\$129,000	\$132,900	\$435,900

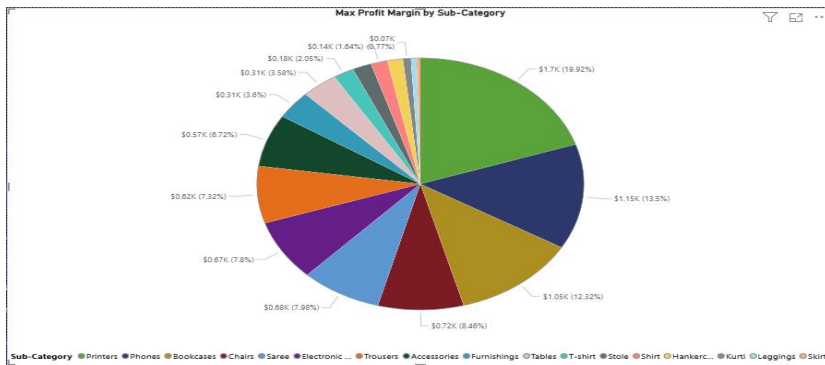
3. Clustered Column Chart

1. Total Sales and Total Target by Category



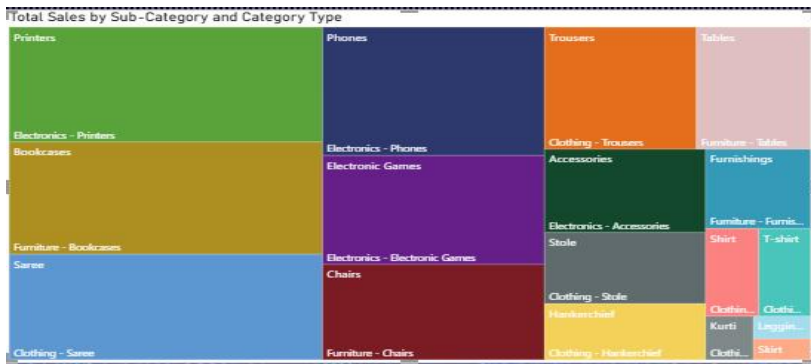
4. Pie Chart

1. Max Profit Margin by Sub-Category



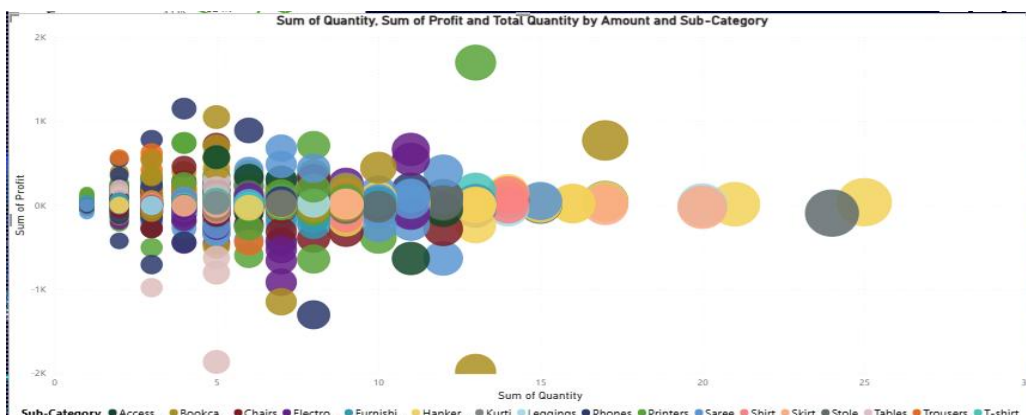
5. Treemap

1. Total Sales by Sub-Category and Category Type



6. Scatter Chart

1. Sum of Quantity, Sum of Profit and Total Quantity Amount by Sub-Category



7. Line Chart

1. Monthly Sales by Month



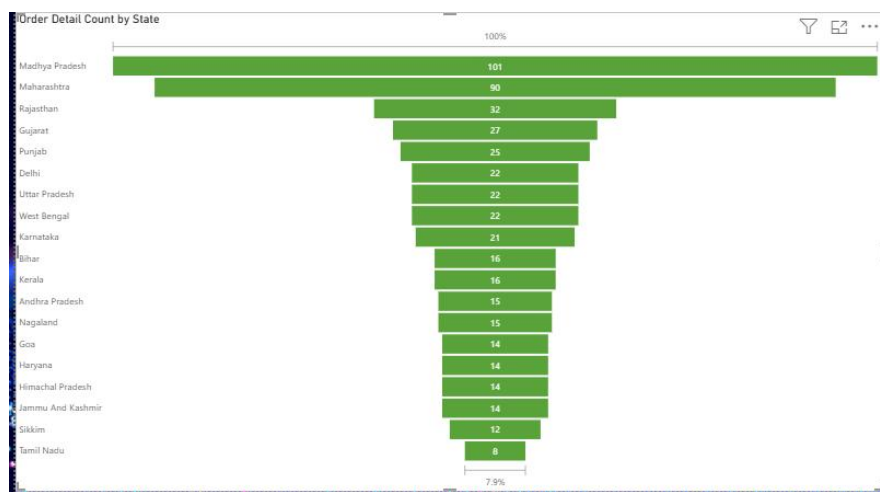
8. Map Visual

1. City Sales by City



9. Funnel Chart

1. Order Detail Count by State





Report / Dashboard Screenshot

This dashboard combines:

- **Descriptive Analysis** → Sales, targets, quantity, profit overview
- **Diagnostic Analysis** → Compare category performance and profit contribution
- **Geographical Analysis** → City-wise sales distribution
- **Trend Analysis** → Monthly sales movement



Analysis

• Descriptive Analysis

Purpose:

Summarize historical business performance.

Findings:

- Total Sales achieved approximately **\$435.9K**
- Target achieved exceeded expectation in several categories.
- Electronics generated the highest sales contribution.
- Monthly sales showed fluctuations across periods.
- Some sub-categories contributed significantly more profit.

Recommendation:

- Continue investment in top-performing categories.
- Monitor low-performing months.

Action Plan:

- ✓ Track monthly KPI dashboard
- ✓ Review category contribution monthly
- ✓ Create executive summary reports

● Diagnostic Analysis

Purpose:

Identify root causes behind performance trends.

Findings:

- Electronics exceeded target due to stronger sales volume.
- Certain sub-categories generated high sales but lower profit margins.
- Geographic distribution indicates concentrated sales regions.

Recommendation:

- Investigate low-profit categories.
- Optimize pricing and promotional strategy.

Action Plan:

- ✓ Conduct category-level profitability review
- ✓ Improve inventory planning
- ✓ Segment customer demand by region

● Predictive Analysis

Purpose:

Forecast future sales and business performance.

Findings:

- Monthly sales trend indicates seasonal fluctuations.
- High-performing categories may continue growth if demand remains stable.
- Current trajectory suggests target achievement is sustainable.

Recommendation:

- Implement sales forecasting models.
- Prepare stock for high-demand categories.

Action Plan:

- ✓ Build forecasting model in Power BI
- ✓ Set automated alerts for declining sales
- ✓ Predict next-quarter demand

● Prescriptive Analysis

Purpose:

Recommend actions to improve future outcomes.

Findings:

- Increase focus on profitable categories.
- Reduce dependency on underperforming products.
- Improve regional sales strategies.

Recommendation:

- Introduce targeted campaigns.
- Optimize pricing and promotions.
- Improve supply chain responsiveness.

Action Plan:

- ✓ Allocate budget to top categories
- ✓ Launch category-specific campaigns
- ✓ Set quarterly target monitoring process

Conclusion – Sales Report Dashboard

The Sales Dashboard provides a comprehensive view of overall business performance across sales, targets, profit, categories, and geographic distribution.

- **Sales Performance:** Total sales reached approximately **\$435.9K**, indicating strong business activity and performance close to or above several target benchmarks.
- **Target Achievement:** Monthly and category-level comparisons show that actual sales performed well against defined targets, with some areas exceeding expectations.
- **Category Insights:** **Electronics** generated the highest sales contribution, followed by **Clothing** and **Furniture**, making Electronics the strongest revenue-driving category.
- **Profit Analysis:** Profit distribution across sub-categories shows that a few product groups contribute significantly more profit than others, highlighting opportunities to focus on high-margin segments.
- **Sub-Category Trends:** Categories such as **Printers, Phones, and Bookcases** contributed noticeably to total sales, while some sub-categories showed lower performance and may require strategic attention.
- **Geographical Performance:** City-wise sales analysis indicates stronger sales concentration in selected regions, suggesting opportunities for targeted regional expansion.
- **Order & Quantity Patterns:** Order count and quantity analysis reveal variations in customer purchasing behavior across categories and locations.